Swisscom Digital Architecture

Bridging the Gap from Telco to TechCo with Agile Architecture

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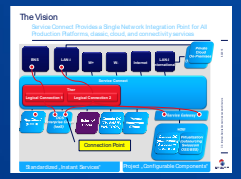
Security Classification | C1 General

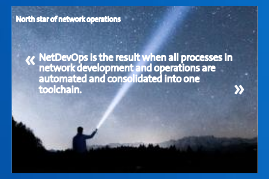
Swisscom's Agile Journey from a Network Perspective

Role of IT has shifted from *Network Fulfilment* to efficient, reliable, and scalable Automation of Operations



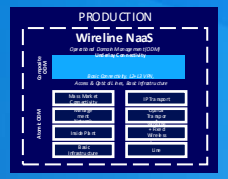
2018

- Merged Dev and Ops into SAFe* org
- First Agile team combining Network & IT



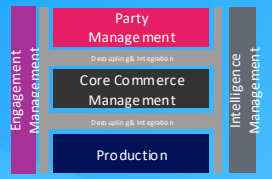
2020

- Split Wireline Network Ops
- Align DevOps and IT Strategy



2021/2022

- New Network IT Strategy
- Successful pilot with IP Networks



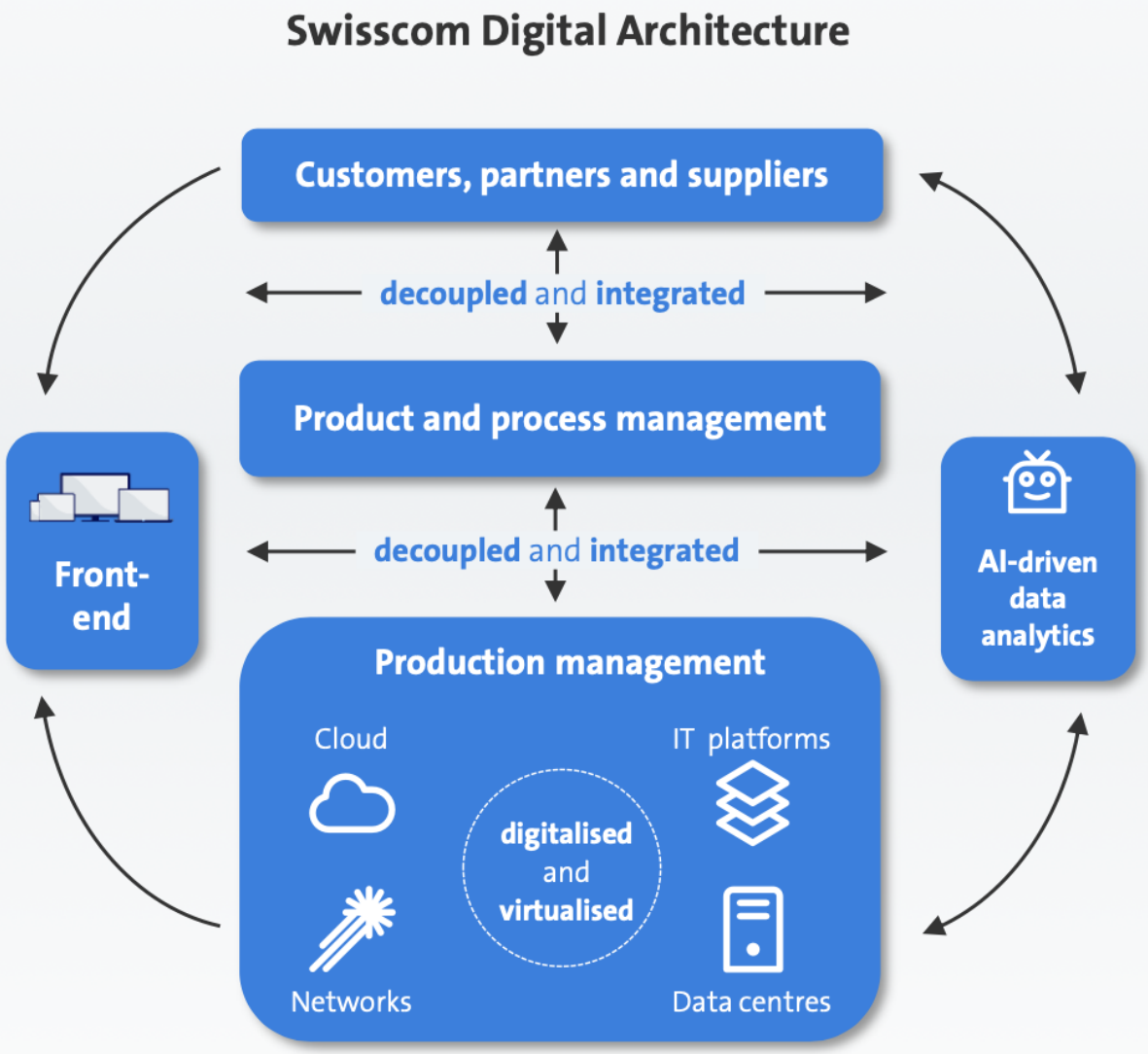
2023/2024

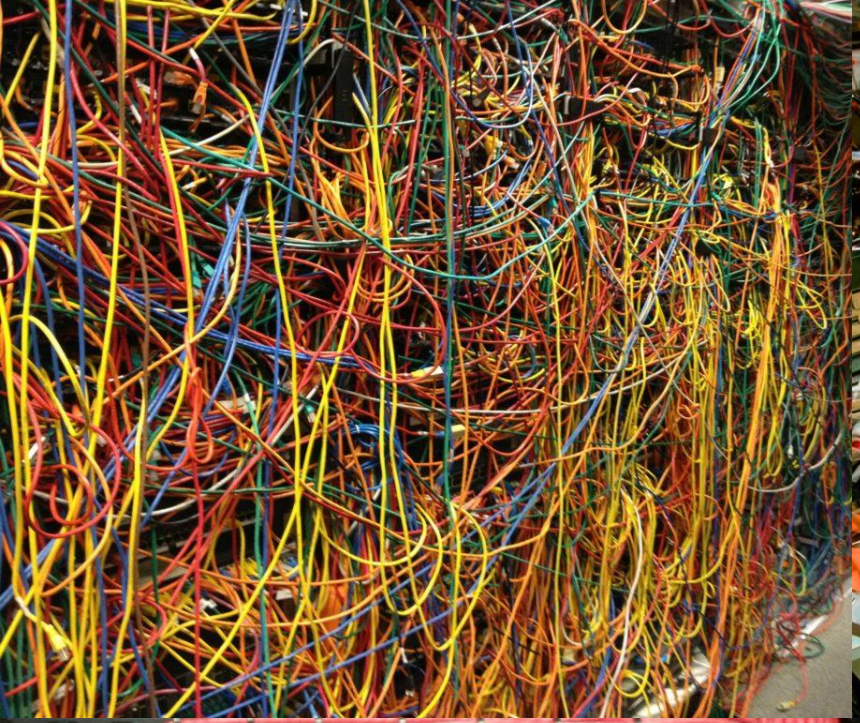
- Kick-off Swisscom Digital Architecture
- Committed company-wide OKRs

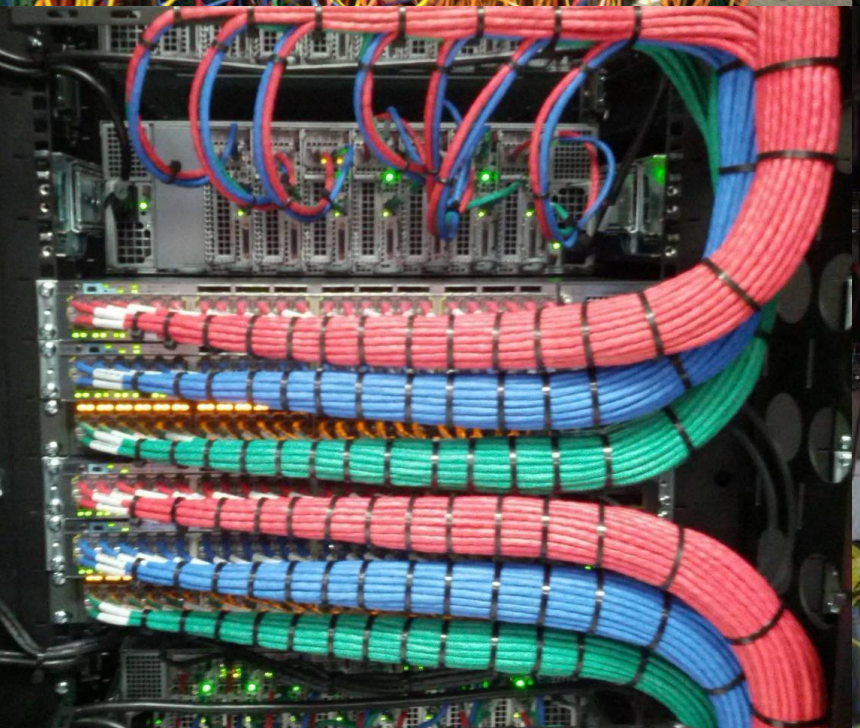
Transforming from Telco to TechCo

Modularize Technological Foundation to increase business autonomy

- Enhance Agility
- Accelerate Time-to-market for new Services
- Improve cost efficiency



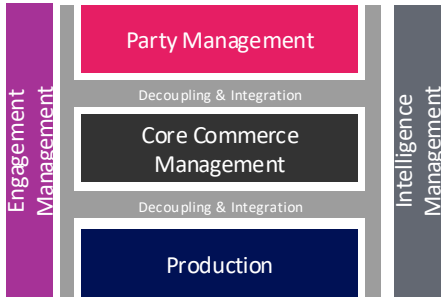


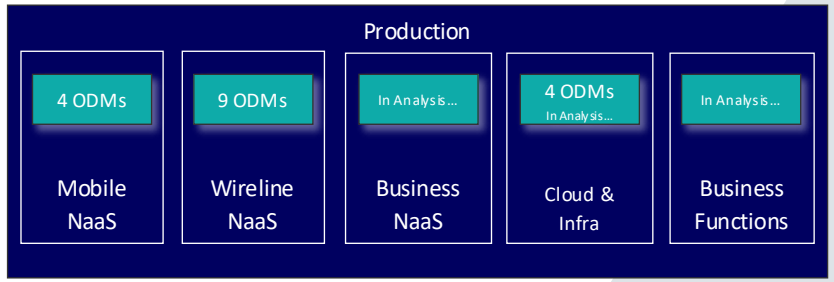


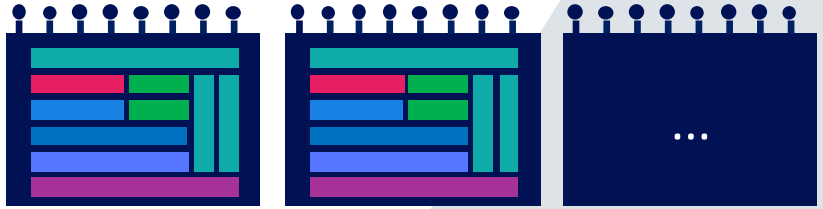
Vacuum
Bike

How to Tame the Beast

Speak a common language, enable orientation across the IT landscape, support coordination and planning







Company Wide

Swisscom Digital Architecture (SDA)

Standards, Frameworks, Compatibility

Value Streams

Operational Domain Management (ODM)

Autonomy, **Dependency Management**, Interoperability

Teams

Technical Architecture Blueprint

Synergies, Roadmaps, IT architecture ownership

From Strategy to Execution: OKR Definition for ODMs

Defined as vector (a, b, c, d) of number of ODMs having passed stage gate

Stage 1: Discover

- Management boundaries
- Resource map
- Services and APIs
- DevOps scoping

Stage 2: Plan

- Technical architecture
- Value hypothesis
- Transition architecture

Stage 3: Execute

- Architecture Runway
- LPM financing
- MVP
- Investment shifted to new IT stack

Stage 4: Operate

- ODM in service
- Standard lifecycle activities

How do we re-architect in autonomous units?

High-level Design & Scoping

ODM	Value Stream	2024	2025	2026
Underlay Connectivity (composite)	VNC	Discover	Execute	Execute
Massmarket Connectivity	VNC	Discover	Plan	Execute
IP Transport	VNC	Execute	Execute	Execute
Optical Transport	VNC		Discover	Plan

Business Planning & Dependencies

		2022 Q1	Q2	Q3	Q4	2023 Q1
Inventories	Ressource inventory decision			■		
	Ressource inventory MVP					■
	Stages, integration, use cases, metro, migration					

Prioritization & Execution

✔	LPMPF-1	Next Generation OSS & Automation Ca
✔	NILS-59	Einführung IP Network Configurat
✔	NILS-42	IPT ODM IT (evolving towards TITA
✔	NILS-33	Lab - Define Lab Strategy

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3 years of experience in Operational Domain IP Transport

Proven sustainable impact on stability, shorter time to market (TTM) and increased efficiency

Reason behind the value	Increase Stability	Shorter TTM	Cost Reduction
Fully automated CI/CD Pipeline for IP Networks : 100% regression testing and staging for each change.	**Reduce 25% of incidents after change** in production compared of today		**80%** of changes can be handled as **SOPs*** during daytime and without TCAB
E2E automated **network configuration management**			**OPEX reduction** (network operations)
Single service definition for IP Transport Service and **standardized APIs** for all processes including fulfillment and assurance		**New Features** are going to be rolled out **within 1 sprint** through reuse of existing generic features	**CAPEX reduction** in development of future network functions
IT is **fully decoupled.** Development and operation fully under control of one organization.	Increase **First Time Right rate**		Reduce **Number of Meetings**
Real-Time Monitoring of all network elements and services with anomaly detection and real-time SLO measurement	**Reduce MTTR by 50%** for IP Transport Network related incidents		

Our Learnings…

Speak a common Language

- Enables Orientation and supports coordination and planning

Re-architect for Agility

- Legacy architecture is a threat to Agile organizations
- Management boundaries must be re-designed for Agile

Decentralize decision making

Re-align global and local decision structures for autonomy